In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2017

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST – I
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**A**	**C**	**B**
2.	**B**	**C**	**C**
3.	**A**	**C**	**B**
4.	**A**	**A**	**C**
5.	**B**	**B**	**B**
6.	**CD**	**ABC**	**AD**
7.	**AB**	**BC**	**CD**
8.	**AD**	**BD**	**AB**
9.	**ABC**	**ACD**	**AD**
10.	**ABD**	**BCD**	**AC**
11.	**BD**	**CD**	**BD**
12.	**ABD**	**AC**	**BCD**
13.	**BD**	**B**	**AB**
1.	**2**	**1**	**3**
2.	**3**	**2**	**8**
3.	**1**	**1**	**5**
4.	**4**	**4**	**0**
5.	**5**	**9**	**8**

Physics

PART – I

SECTION – A

1. A

Sol. For C $5mg - 3kx_3 = 0 \quad \Rightarrow 3kx_3 = 5mg$

For B $3kx_3 + 2mg = 2k.x_2 \Rightarrow 2kx_2 = 7mg$

For A $2kx_2 + 3mg = kx_1$

If spring 2 is cut $kx_2 = 0$

$2mg + 3kx_3 = 2ma_B$

$7\ mg = 2ma_B$

$a_B = \frac{7}{2}g(\downarrow)$

2. B

Sol. $\vec{v} = 2V \cos 30° \sin 30° (-\hat{i}) + V(\cos^2 30° - \sin^2 30°)\hat{j}$

3. A

Sol. $\lambda_{min} = \frac{12375}{40,000} = 0.30$ Å. Hence wavelength less than 0.30 Å is not possible.

4. A

Sol. Potential energy $U = eV = eV_0 \ln\frac{r}{r_0}$

$\therefore$ Force $F = \left|\frac{dU}{dt}\right| = \frac{eV_0}{r}$.

$\therefore$ The force will provide the necessary centripetal force.

Hence, $\frac{mv^2}{r} = \frac{eV_0}{r} \Rightarrow v = \sqrt{\frac{eV_0}{m}}$ …(i)

and $mvr = \frac{nh}{2\pi}$ …(ii)

From equations (i) and (ii), $mr = \left(\frac{nh}{2\pi}\right)\sqrt{\frac{m}{eV_0}}$ or $r \propto n$.

5. B

Sol. $y = A \sin(kx) \cos(\omega t)$

$k = \frac{2\pi}{\lambda} = \frac{\pi}{\ell}$

$\ell \rightarrow$ length of sting

$U = \int dU$

$= \int_{x=0}^{x=\ell} \frac{1}{2}(dm)\,\omega^2 A^2 \sin^2\left(\frac{\pi x}{\ell}\right)$ and $dm = \frac{m}{\ell}dx$

6. CD

Sol. Particle *P* will have a velocity in vertical direction.

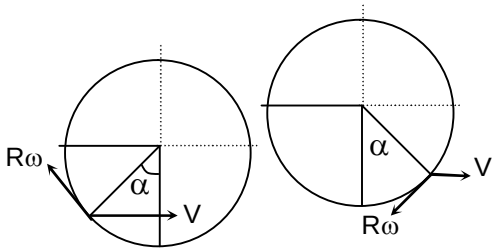
i.e. $R\,\omega\cos\alpha = V$

$$\alpha = \cos^{-1}\frac{V}{R\omega}$$

Required angle θ is $\theta = \pi - \alpha = \pi - \cos^{-1}\frac{V}{R\omega}$

And in second case $R\omega\cos\alpha = V$

$\alpha = \cos^{-1}\frac{V}{R\omega}$, $\theta = \pi + \cos^{-1}\frac{V}{R\omega}$



7. AB

Sol. $-\frac{dv}{dt} = \frac{v^2}{R}$ & $-v\frac{dv}{ds} = \frac{v^2}{R}$

8. AD

Sol. Property of friction.

9. ABC

Sol. Work energy theorem.

10. ABD

Sol. By conservation of angular momentum,

$I_1\omega_i = (I_1 + I_2)\omega_f$

$\omega_f = \omega_1/3$

$E_i = \frac{1}{2}I_1\omega_i^2$

$E_f = \frac{1}{2}(I_1 + I_2)\omega_f^2$

Ratio of the heat produced to initial kinetic energy $= \frac{E_i - E_f}{E_i} = \frac{2}{3}$

11. BD

Sol. $E \propto m$ and $r \propto \frac{1}{m}$

12. ABD

Sol. Bernoulli's theorem for an orifice at depth 'x' in liquid '3d'.

$P_o + \left(dg\frac{H}{2} + 3dg.x \right) = P_o + \frac{1}{2} \times 3d \times v^z$ …(1)

$\frac{H}{2} - x = \frac{1}{2}gt^2$ …(2)

$R = Vt$ …(3)

Solve for R and apply maxima/minima.

13. BD

Sol. Stopping potential does not depend on intensity and saturation current depends on intensity. Intensity I due to point source $\propto \frac{1}{r^2}$ where r is distance from the source.

SECTION – C

1. 2

Sol. For m not to slide over M, acceleration of the triangular block should be g $\tan\theta$ if the required mass in m_0, then

$m_0 g - T = m_0 (g\tan\theta)$

& $T = (M + m) g\tan\theta$ &

$T = 2g\tan\theta$

& $\cot\theta = 2$

$m_0 = 2$ kg

2. 3

Sol. $PT^2 = k$

$\gamma = \frac{1}{V}\left(\frac{dV}{dT}\right)$ …(i)

$\left(\frac{nRT}{V}\right) T^2 = k$

$\Rightarrow \frac{T^3}{V} = k'$

$\Rightarrow V = \frac{T^3}{k'}$ …(ii)

$\frac{dV}{dT} = \frac{3T^2}{K'}$ …(iii)

From (i), (ii) and (iii)

$\gamma = \left(\frac{k'}{T^3}\right) \times \left(\frac{3T^2}{k'}\right) = \frac{3}{T}$

3. 1

Sol. $B\int_0^R \frac{\omega}{2\pi} Q \frac{2\pi r}{\pi R^2} dr\, \pi r^2 = \frac{T_0 D}{2}$

$\Rightarrow \omega = \frac{T_0 D}{QR^2 B}.$

4. 4

Sol. Consider energy losses in successive collisions, find θ, i.e. maximum deflection after each collision you will get $\cos\theta > \frac{1}{2}$ will satisfy $\theta < 60^0$ this condition will be met after nth collision.

After 1st collision, K.E. = $mghe^2$ = 0.8 mgh

After 2nd collision, K.E. = 0.64 mgh

After 3rd collision, K.E. = 0.512 mgh

After 4th collision, K.E. < $\frac{mgh}{2}$

i.e., $\theta < 60°$.

5. 5

Sol. $5 = \frac{i}{2}(6) + i(2) + iR$

$\Rightarrow \quad 5 = 3i + 2i + iR$

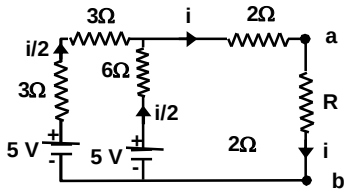
$\Rightarrow \quad i = \frac{5}{5+R}$

$P = i^2R = \left(\frac{5}{5+R}\right)^2 R$

$P = i^2R$

$\frac{dP}{dR} = 0$

$R = 5\Omega.$



Chemistry

PART – II

SECTION – A

1. C

Sol. $CaCO_3 + H_2SO_4 \longrightarrow CaSO_4 + CO_2 + H_2O$

Moles of $CaCO_3$ = Moles of H_2SO_4 = $\frac{25.1 \times \frac{0.02}{2}}{1000} = 2.51 \times 10^{-4}$

Moles of $CaCO_3 = 2.51 \times 10^{-4} \times 100 = 2.51 \times 10^{-2}$

100 mL water contains = 2.51×10^{-2} g $CaCO_3$

$\therefore 10^6$ mL(10^6 g) water will contain = 251 g of $CaCO_3$

$\therefore$ Hardness of sample = 251 ppm

2. C

Sol. NaOH is not used as a bleaching agent.

3. C

Sol.

	$A(g)$	$\rightarrow$	$4B(g)$	+	$3C(g)$
At $t=0$:	P_o		0		0
At $t=20\,min$:	$P_o - x$		4x		3x

$P_o - x + 4x + 3x = 400 \Rightarrow P_o + 6x = 400 \Rightarrow 6x = 400 - 100 = 300$

x = 50

$K = \frac{0.693}{t_{1/2}} = \frac{0.693}{20 \times 60} = 5.7 \times 10^{-4}\ sec^{-1}$

4. A

Sol. The acidic strength of the acids is

HW > HX > HY > HZ

5. B

Sol. No. of X^{2+} ions per unit cell = 1

No. of Y^{2-} ions per unit cell = $8\,Y^{2-} \times \frac{1}{8} = Y^{2-}$

Hence the formula is XY.

6. ABC

Sol. Dipole moment of $NH_3 > NF_3$, $SCl_2 > BeCl_2$(zero value)

$SF_4 > CF_4$(zero value)

7. BC

Sol. It is an endothermic reaction as the equilibrium constant increases with increasing temperature.

8. BD

Sol. $Be_2C \xrightarrow{H_2O} Be(OH)_2 + CH_4$

$Al_4C_3 \xrightarrow{H_2O} Al(OH)_3 + CH_4$

9. ACD

Sol. $\Delta S_{x\to z} = \Delta S_{x\to y} + \Delta S_{y\to z}$ (Enthalpy is a state function and hence additive)

$\Delta H_{x\to y\to z} = \Delta H_{x\to z}$ (State function, depend in initial and final state)

$W_{x\to y\to z} = W_{x\to y}$ (work done in $y \to z$ is zero as it is an isochoric process)

10. BCD

Sol. Gr-III radicals are precipitated with ammonia solution.

11. CD

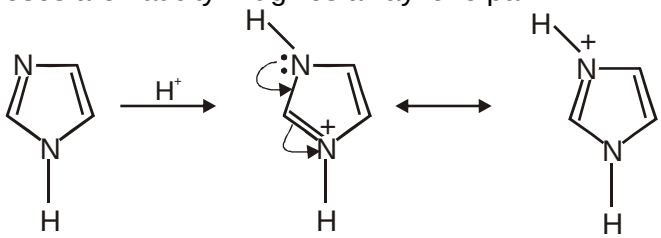
Sol. For Ni^{2+} the configuration is $t_{2g}^6 e_g^2$ and Zn^{2+} is $t_{2g}^6 e_g^4$.

12. AC

Sol. $\mu = 2\mu\cos\frac{\theta}{2}$ [When two diploes are identical]

13. B

Sol. (i) Loses aromaticity if it gives away lone pair

(ii) 

Conjugate acid is resonance stabilized but attached to sp^2 carbon atoms, which shows – I effect and destabilize the conjugate acid.

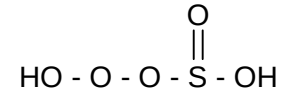
SECTION – C

1. 1

Sol. The valence electrons are $3s^2 3p^1$ for which ℓ = 0 and 1.

2. 2

Sol. The structure of H_2SO_5 is:



3. 1

Sol. Total charge on four oxide ions = $-2 \times 4 = -8$

Let 'n' no. of $X^{-8/3}$ ions will replace O^{2-} ions

$\therefore\ n \times \frac{-8}{3} = -8 \Rightarrow n = 3$

$\therefore$ One vacancy will be created.

4. 4

Sol. $\Delta T_f = K_f \times n \times \frac{1000}{W}$

Or, $4 = 1.86 \times \frac{6.2}{62} \times \frac{1000}{W}$

$\therefore$ W = 46.5 g

$\therefore$ Mass of water converted to ice = 50.5 – 46.5 = 4 g

5. 9

Sol. Charge = $96500 \times 1 = 96500$ coulomb = 1 F

1F deposits 1 g eq = 9 g of aluminium

Mathematics

PART – III

SECTION – A

1. B

Sol. $|A+B|=0$

2. C

Sol. 4 circles touch sides of any given triangle formed by 3 tangents.

3. B

Sol. Required number $=8\times9\times9\times9\times3$

4. C

Sol. Let common root be α and $P(x)=x^2+bx+c$

Now $P(\alpha)=0$ and $P(P(P(\alpha)))=0$

$\therefore$ one of the roots must be 1.

5. B

Sol. $(x+iy)^3=18+26i$

$\Rightarrow x^3-3xy^2=18$

$3x^2y-y^3=26$

Use $y=tx$ and solve

6. AD

Sol. $f'(x)=\lim\limits_{n\to0}\dfrac{f(x+h)-f(x)}{h}=|x|$

Hence, f (x) is differentiable for all $x\in R$ but not twice differentiable at x = 0.

7. CD

Sol. $x\ell n2<F(x)<x^2\ell n2$ as $x\to1^+$

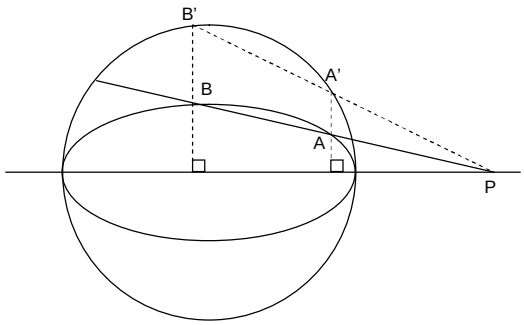
$\therefore$ Range of $F(x)$ is $(\ell n2,\infty)$

8. AB

Sol. Consider a number system with base 3 and using only the digits 0 or 1.

9. AD

Sol. Take $A(\theta_1)$ and $B(\theta_2)$



10. AC

Sol. Second diagonal will pass through the centre of the hyperbola.

11. BD

Sol. $2978^n - 2978^m = 1000k \quad k \in N$

$\therefore m = 3\lambda$ and $n - m = 100\mu$

12. BCD

Sol. Let $\vec{r_1} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\vec{r_2} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\left|\vec{r_1} \times \vec{r_2}\right| \le \left|\vec{r_1}\right|\left|\vec{r_2}\right|$

13. AB

Sol. Radical axis of $S_1 = 0$ and $S_2 = 0$ is the line $L_1 = 0$, the point of concurrency of family of lines $ax + by + c = 0$ if a,b,c are in A.P. is (1,-2)

SECTION – C

1. 3

Sol. $g(1) = e^2$

$$\lim_{x \to 1^-} g(x) = \frac{h(1) + 1}{6}$$

$$\lim_{x \to 1^+} g(x) = \frac{f(1)}{2}$$

Since, $g(x)$ is continuous at $x = 1$

$$g(1) = \lim_{x \to 1^-} g(x) = \lim_{x \to 1^+} g(x)$$

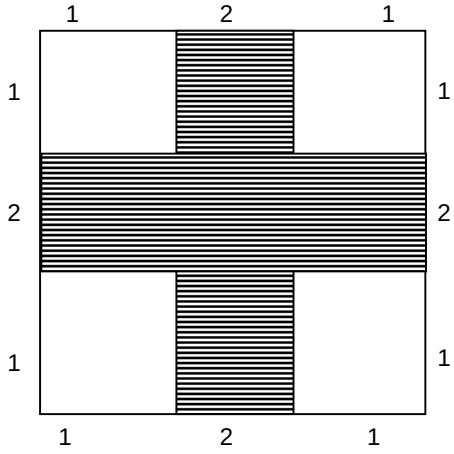
$$\Rightarrow 2g(1) + 2f(1) - h(1) + 2 = 3$$

2. 8

Sol. $\text{Area} = \int_0^1 \left(6 - f(x)\right)dx + \int_{-1}^0 \left(f(x) - (-2)\right)dx = \frac{5}{4}$

3. 5

Sol. Feasible region is



4. 0

Sol. Focus is $\left(\frac{2}{5}, -\frac{4}{5}\right)$

5. 8

Sol. $x^4 - 7x^2 - 4x + 20 = \left(x^2 - 4\right)^2 + \left(x - 2\right)^2$

$x^4 + 9x^2 + 16 = \left(x^2 + 4\right)^2 + x^2$

Take the curve $y = x^2$.

Both square roots can be interpreted as distances.